

Separating the completeness budget from its spatial placement:

the aggregate- C completion in darksirens

Technical note accompanying PR #338 (feat/completeness-aggregate-budget)

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August 8, 2026

Abstract

The dark-siren redshift prior completes a magnitude-limited galaxy catalog with a missing-galaxy branch whose differential completeness C was estimated per HEALPix pixel from the observed counts themselves. Because the estimator divides a per-pixel numerator by an isotropic expectation, it measures the product of the survey selection and the local galaxy overdensity, $C \simeq C_{\text{sel}}(z) [1 + \delta_{\text{obs}}(p, z)]$, clipped to $[0, 1]$: angular clustering is absorbed into the “how much is missing” budget, and the predicted missing density *anti-correlates* with the true one. This note derives the failure, documents the corrected estimators introduced in PR #338 — a sky-aggregate budget $\tilde{C}(z)$ that reserves all spatial structure for the completion field $Q_{\text{LSS}}(p, z)$, a per-redshift mean-one renormalization that makes the missing budget invariant to Q -fit noise by construction, a matched truncation of the expected counts at the survey depth, and a resolution guard on the low-rank Gaussian-process completion — and quantifies every change on a closure experiment with known injected clustering and on an end-to-end H_0 scan of 300 mock dark sirens. All legacy behavior is preserved bit-for-bit at default settings; the new estimators are opt-in and provenance-locked.

1 Setup and notation

Let z_g denote the package redshift grid (N_g log-spaced nodes on $[0, z_{\text{max}}]$) and $p = 1, \dots, N_{\text{pix}}$ the HEALPix pixels of solid angle $a_{\text{pix}} = 4\pi/N_{\text{pix}}$. The homogeneous expectation for the observable galaxy density per pixel per unit redshift is

$$dN^{\text{exp}}(z) = n_0 a_{\text{pix}} \frac{dV_c}{dz d\Omega}(z) (1+z)^\delta, \quad (1)$$

with n_0 the comoving density and δ a phenomenological evolution index; both are sampled in the hierarchical likelihood. Every density that enters a ratio is smoothed by one linear operator S , built from unit-mass truncated Gaussian kernels of width $\sigma_s = 0.05$: the observed-galaxy KDE in pixel p is

$$dN_s^{\text{obs}}(z|p) = \sum_{i \in p} \frac{\mathcal{N}(z; z_i, \sigma_s)}{M(z_i)}, \quad M(z_i) = \Phi\left(\frac{z_{\text{max}} - z_i}{\sigma_s}\right) - \Phi\left(\frac{-z_i}{\sigma_s}\right), \quad (2)$$

and the matched expectation is $dN_s^{\text{exp}} = S dN^{\text{exp}}$. Because S is shared and linear, a redshift-independent true completeness passes through the ratio exactly.

The assembled per-pixel redshift prior is the in/out mixture

$$p(z|p) = \frac{N_{\text{obs}}(p) p_{\text{cat}}(z|p) + dN^{\text{miss}}(z|p)}{N_{\text{obs}}(p) + N_{\text{miss}}(p)}, \quad N_{\text{miss}}(p) = \int dN^{\text{miss}}(z|p) dz, \quad (3)$$

with p_{cat} the unit-normalized catalog kernel density and N_{obs} the observed count. In *conditional* sky weighting every pixel carries unit mass; in *field* weighting the denominator is replaced by the survey-global $Z = \sum_p [N_{\text{obs}}(p) + N_{\text{miss}}(p)]$, restoring relative angular host density. The missing branch is

$$dN^{\text{miss}}(z|p) = [1 - C(z|p)] dN^{\text{exp}}(z) \ell(p, z), \quad (4)$$

where the large-scale-structure factor ℓ is exactly one of

$$\ell(p, z) = \begin{cases} 1 & \text{homogeneous,} \\ \max(1 + b_{\text{eff}} \delta_g(p, z), 0) & \text{legacy local overdensity,} \\ Q_{\text{LSS}}(p, z) = e^{\log Q(p, z)} & \text{precomputed completion field,} \end{cases} \quad (5)$$

the latter two mutually exclusive. Beyond a declared survey depth z_{depth} the convention is $C \equiv 0$, $\ell \equiv 1$ (hosts are missing, not nonexistent).

2 The legacy per-pixel estimator and its failure modes

2.1 Clustering absorption (the thinning degeneracy)

The legacy differential completeness is estimated per pixel,

$$C_{\text{pp}}(z|p) = \text{clip}\left(\frac{dN_s^{\text{obs}}(z|p)}{dN_s^{\text{exp}}(z)}, 0, 1\right). \quad (6)$$

The numerator is per-pixel data; the denominator is isotropic. Writing the observed density as selection times local structure, $dN_s^{\text{obs}} \simeq S [C_{\text{sel}} dN^{\text{exp}} (1 + \delta_{\text{obs}})]$, the estimator returns

$$C_{\text{pp}}(z|p) \simeq \text{clip}(C_{\text{sel}}(z) [1 + \delta_{\text{obs}}(p, z)], 0, 1) : \quad (7)$$

the angular clustering of the *observed* galaxies is absorbed into the completeness. This is the classical thinning (selection-density) degeneracy of an inhomogeneous Poisson process: from counts alone only the product $\lambda = \phi \cdot n$ is identifiable, and splitting it requires external structure. Galaxy surveys avoid counts-based completeness for precisely this reason (injection campaigns and survey-property templates are constructed to be independent of the real density field); among dark-siren pipelines the counts-based per-pixel construction appears in the DarkSirensStat mask/cone completeness, with the absorption pathology acknowledged there.

Two immediate consequences follow from Eq. (7). First, an overdense *complete* pixel reads $C > 1$ and is clipped: its excess is silently discarded. Second, and structurally worse, the missing branch anti-tracks the truth: to first order in δ ,

$$1 - C_{\text{pp}} \simeq (1 - C_{\text{sel}}) - C_{\text{sel}} \delta_{\text{obs}}, \quad (8)$$

so a pixel with *more* true (and hence more missing) galaxies is assigned *fewer* predicted missing galaxies. On the closure mock (Sec. 4) the measured anomaly correlation of $\log dN^{\text{miss}}$ against truth is $r = -0.43$ with slope -0.62 . In the assembled prior (3) the observed and missing branches then largely cancel the catalog's clustering (total-density slope 0.25 against truth), and because the completion field Q_{LSS} is fit to the *same* counts residual to C_{pp} (its Poisson base is $C_{\text{pp}} dN^{\text{exp}}$), it can only ever recover sub-smoothing structure — measured slope 0.29, against 0.77 for the crude legacy δ_g factor.

2.2 Jensen inflation of the missing budget

The completion tables store the Laplace posterior mean $\mathbb{E}[Q] = \exp(b s_{\text{MAP}} - \text{shift} + \frac{1}{2} b^2 \sigma_{\text{post}}^2)$. The posterior variance σ_{post}^2 is largest exactly where data are sparse, so the spatially varying lognormal factor $e^{\sigma^2/2}$ does not integrate away: on the mock the radial-mode missing budget is inflated by +55% relative to the homogeneous budget. The budget does not belong to Q at all — it is the job of C and the sampled n_0 .

2.3 Resolution collapse of the low-rank completion field

The 3-D completion field is a low-rank Gaussian process on (sphere $\times \zeta = \log(1+z)$) with M_z inducing nodes spanning $[0, \zeta_{\text{hi}}]$. The shipped hyperparameters fixed the correlation length at 50 Mpc, which maps to $\ell_\zeta = L(d\zeta/d\chi)|_{z_{\text{ref}}} \approx 0.01$, roughly $30\times$ below the node spacing $\zeta_{\text{hi}}/(M_z - 1) \approx 0.36$. In this regime a variational/low-rank GP posterior collapses toward its prior while the optimizer converges normally (Burt et al. 2019): the fitted field is nearly blind to real radial structure (measured fitted-vs-truth $\log Q$ slope 0.04), with no configuration error raised and no CLI knob available to change the lengthscale.

2.4 Depth-edge mismatch

The galaxy kernels (2) renormalize on $[0, z_{\text{max}}]$ of the grid, while the expected side smoothed the *untruncated* dN^{exp} . Just below a declared depth z_{depth} the numerator loses the kernel mass beyond the edge but the denominator does not, so C dips and $dN^{\text{miss}} = (1 - C) dN^{\text{exp}}$ spikes within $\sim 2\sigma_s$ of the edge.

3 The corrected estimators (PR #338)

3.1 S0a: matched truncation at the survey depth

With a concrete z_{depth} the expected side is truncated identically to the observed side before smoothing,

$$dN_s^{\text{exp}}(z) = [S(dN^{\text{exp}} \cdot \mathbf{1}[z \leq z_{\text{depth}}])](z), \quad (9)$$

so a constant true completeness again passes through the ratio exactly at the edge (the same construction the $z = 0$ boundary already used). Beyond the depth the denominator is pinned to an inert constant: the truncated smooth underflows there, and a live data/ ϵ division feeds $0 \times \infty$ NaN cotangents through the downstream `clip/where` in reverse-mode differentiation even though every consumer discards C beyond the depth. The $z_{\text{depth}} = \text{None}$ path is bit-identical to the legacy expression.

3.2 S0b: per-redshift mean-one budget renormalization

After any table build (radial or gp3d), with build-time weights $w_p(z) = [1 - C_p(z)] dN^{\text{exp}}(z)$ over the fitted footprint $\mathcal{F}$,

$$\log Q_p(z) \longleftarrow \log Q_p(z) - \log \frac{\sum_{p' \in \mathcal{F}} w_{p'}(z) Q_{p'}(z)}{\sum_{p' \in \mathcal{F}} w_{p'}(z)}, \quad (10)$$

applied to the MAP table *and independently to every Laplace ensemble member*. The identity $\sum_p w_p Q_p = \sum_p w_p$ then holds exactly per redshift bin: Q becomes a pure *redistribution* field, the

missing budget with any Q equals the homogeneous budget by construction (killing the +55% Jensen inflation identically), and the ensemble carries placement uncertainty but zero budget uncertainty — the budget lives in C and the sampled n_0 , where it is marginalized. The removed monopole curve and a boolean stamp are recorded in the HDF5 provenance (fail-closed writer; loud loader warning for unstamped legacy tables). The invariant is defined over the *fitted* footprint: occupied pixels for the per-pixel base, the whole sky for the aggregate base; the gp3d “borrowing halo” evaluated onto unfitted pixels deliberately sits outside it.

3.3 S0c: hyperparameter exposure and the resolution guard

The builder gains `--lss-corr-length-mpc`, `--lss-sigma`, `--gp3d-nz-nodes`, `--gp3d-nsph-nodes`, and `--gp3d-z-node-hi` (default moved from a hardwired $\zeta_{\text{hi}} = \log 4$ to the grid maximum). A *hard error* fires at build time when

$$\frac{\zeta_{\text{hi}}}{M_z - 1} > \ell_\zeta, \quad (11)$$

stating both numbers and the minimum admissible M_z ; the sphere side warns when $\sqrt{4\pi/M_{\text{sph}}} > \ell_{\text{sph}}$. The configuration that produced the slope-0.04 collapse can no longer build silently.

3.4 S1: the aggregate budget $\bar{C}$

The opt-in mode `c_mode="aggregate"` replaces the per-pixel ratio (6) by one sky-aggregate curve per likelihood proposal,

$$\bar{C}(z) = \text{clip} \left(\frac{\sum_p dN_s^{\text{obs}}(z|p)}{N_{\text{pix}}^{\text{tot}} dN_s^{\text{exp}}(z)}, 0, 1 \right), \quad N_{\text{pix}}^{\text{tot}} = \text{round} \left(\frac{4\pi}{a_{\text{pix}}} \right), \quad (12)$$

broadcast identically to every pixel (occupied *and* empty) in both the conditional and field paths, so numerator and global normalizer carry the same budget by construction. The numerator is a θ -independent data constant (reusing the existing full-sky KDE rows; a compact catalog without them is refused rather than silently under-counted); all parameter dependence enters through the shared denominator. Averaged over the footprint the density fluctuations cancel, $\sum_p \delta_p \approx 0$, so $\bar{C}$ estimates the *selection alone*: on the mock it matches the analytic generative selection to 0.3% in median absolute deviation. The missing branch becomes

$$dN^{\text{miss}}(z|p) = [1 - \bar{C}(z)] dN^{\text{exp}}(z) \ell(p, z), \quad (13)$$

with $(1 - \bar{C})$ pixel-independent: the entire spatial signal is carried by ℓ , whose correlation with the truth flips positive by construction. Correspondingly the completion-field fit conditions on the base $\bar{C} dN^{\text{exp}}$ and *includes empty pixels* as $N_{\text{obs}} = 0$ rows against the nonzero base — the zero count is the void information the occupied-only fit discarded. Q now targets the full observed overdensity rather than the sub-smoothing residual. Tables record their `c_mode` in provenance and consumption is fail-closed in both directions (the two targets differ by the whole clustering signal; mixing them double-counts or drops it).

Two documented caveats. $\bar{C} \propto 1/n_0$ through a single global clip, so the likelihood can be kinked in the sampled n_0 near full completeness. And the sky-aggregate assumption presumes isotropic survey depth; for genuinely anisotropic compilations the sanctioned extensions are stratified $\bar{C}_k$ per depth region or a magnitude-based $C_{\text{sel}}(z; \theta)$ fit from the per-galaxy magnitude

likelihood (which is independent of the density field by thinning) — staged as future work, not part of this PR.

3.5 Auxiliary fix: grid endpoint vs. distance table

Under a non-default DARKSIRENS.ZMAX, XLA’s and NumPy’s `expm1 ◦ log` can disagree by one ulp (e.g. at $z_{\max} = 0.52$), which left the redshift grid’s last node *above* the NumPy-built distance table; the out-of-range mask then returned NaN for $r(z_{\max})$ and poisoned every consumer of $\chi(z_g)$ (measured: an all-NaN radial Q table, refused by the fail-closed writer). The endpoint is now clamped to the NumPy value when lower — bit-inert at the default $z_{\max} = 5$.

4 Validation

4.1 Closure experiment (known injected clustering)

Mean-one lognormal truth field from the same low-rank GP family the builder fits (closure exact in principle), fiducial survey selection, 277k galaxies, 800 Mpc correlation length, **nside** 16. Both modes rebuilt with resolved nodes ($M_z = 12$ to $z = 0.5$, $M_{\text{sph}} = 64$); the per-pixel column isolates the S0-only fixes (its anti-correlation is the structural defect S1 removes).

	per-pixel C (legacy + S0)	aggregate $\bar{C}$
missing-branch spatial closure (r , slope)	−0.44, −0.64	+ 0.84 to + 0.85 , 0.81–0.99
assembled total-density slope (gp3d / radial / δ_g)	0.27 / 0.26 / 0.75	0.92 / 0.99 / 1.07
fitted-vs-truth log Q slope (gp3d)	0.26 (0.04 shipped)	0.80
void voxels (true $Q \leq 0.25$): median fitted Q	0.94–0.95	0.32–0.46
Q missing budget vs. homogeneous	radial +55% → equal	equal (by construction)
depth-edge dN^{miss} spike	removed (S0a)	removed
aggregate C vs. generative selection	0.3%	0.3%

Table 1: Closure metrics before/after. Bold: the S1 architecture change. The +8% residual budget offset shared by *all* configurations (including homogeneous) is the realization’s sample-variance monopole ($\langle Q_{\text{truth}} \rangle_{\text{sky}} \approx 0.89$ near depth) and belongs to the sampled n_0 .

4.2 End-to-end H_0 scan

300 GW dark sirens hosted by the clustered catalog with a DAG-consistent pipeline (detection cut on the observed SNR; the PE conditions on the same recorded noise realization; 2×10^6 selection injections share the statistic and threshold and are reweighted across trial H_0 ; selection $N_{\text{eff}} > 12,000$ across the whole grid). Machinery closure gates passed before reading results: a fully generator-native mock recovers $H_0 = 67.0$ (truth 67.74), and the complete-catalog gate on the clustered events gives 69.5 ± 1.4 (+1.3 σ).

5 Compatibility and provenance summary

- Legacy per-pixel mode remains the **default** and bit-identical (pinned by `test_z_depth_none_bit_identical` and `test_per_pixel_default_bit_identical`); `c.mode` is structural at trace time (a leaf-less sentinel), and a jit-traced integer flag hard-errors rather than silently selecting a mode.

configuration	per-pixel C		aggregate $\bar{C}$	
homogeneous	69.07 ± 1.38	$(+1.0\sigma)$	67.87 ± 1.32	$(+0.1\sigma)$
legacy $1 + b\delta_g$	68.07 ± 1.35	$(+0.2\sigma)$	66.80 ± 1.39	(-0.7σ)
Q radial	69.11 ± 1.38	$(+1.0\sigma)$	66.91 ± 1.37	(-0.6σ)
Q gp3d	68.99 ± 1.37	$(+0.9\sigma)$	67.00 ± 1.36	(-0.5σ)

Table 2: Gaussian peak fits to the fine-grid $\ln \mathcal{L}(H_0)$ scans (truth $H_0 = 67.74$; survey block fixed at truth). Every configuration recovers truth within $\sim 1\sigma$ at $n = 300$: the ensemble H_0 is statistics-dominated at this depth. Because all scans share the same events, the $\sim 1.2 \text{ km s}^{-1} \text{ Mpc}^{-1}$ coherent offset between the per-pixel and aggregate families is nearly noise-free *on this realization* but is a single draw; certifying it as a method bias requires coverage over disjoint realizations. At $n \sim 3000$ ($\sigma \approx 0.45$) the same shift would be a $3\text{--}4\sigma$ systematic.

- Q tables are stamped with `c_mode` and the budget-renormalization state; consumption fails on any mismatch, and unstamped legacy tables load as per-pixel with a loud warning.
- Deferred, in order: mixture-of-Laplace members over a marginal-likelihood hyperparameter grid (member coverage is still overconfident: $5\% \rightarrow 19\%$); post-renormalization ± 7 clip-railing accounting; `--validate_completion` under aggregate mode; stratified or magnitude-based C_{sel} for anisotropic-depth catalogs, with the selection parameters sampled inside the likelihood and entering the selection integral.